

Question	Marking Guidance	Mark	Comments
08.1	1. 2 nuclei (in cells) OR Cells (stopped) at telophase; 2. Cytokinesis prevented OR Stopped (new) cell membrane forming OR Stopped cytoplasm dividing;	2 (2 x AO3)	2. Accept cell membrane not dividing/splitting/ pinching (in the cell)

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08.2	1. (MiTMAB) binds (to dynamin) other than the active site; 2. Changes the shape of (dynamin) active site OR Changes the tertiary structure (of dynamin/ enzyme); 3. Not <u>complementary</u> so substrate does not bind (to active site) OR Not <u>complementary</u> so no/fewer enzyme-substrate complexes (form);	3 (1 x AO1, 2 x AO2)	1. Accept (MiTMAB) binds to dynamin at an allosteric OR inhibitor site 2. Accept denature for 'change in shape' 3. Accept ES complex in this instance 3. Ignore ESC 3. Accept fit OR attach for bind

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08.3	1. (At) lowest concentrations (all) dynamin is not inhibited OR (At) lowest concentrations (MiTMAB) does not cause cell death/inhibit cytokinesis; 2. (As MiTMAB) concentration increases more dynamin is inhibited/inactive OR (As MiTMAB) concentration increases cell death increases OR (As MiTMAB) concentration increases cytokinesis decreases OR No change in cell number at 2000 ($\mu\text{g dm}^{-3}$) OR No change in cell number at 0.0 (on y axis); 3. (At) highest (MiTMAB) concentrations all dynamin is inhibited OR (At) highest concentrations (MiTMAB) causes cell death OR (At) highest concentrations (MiTMAB) inhibits cytokinesis;	3 (3 x AO2)	1. Accept graph readings in range 30 to 70 for lowest; 1. Accept 'has no effect' for 'cause cell death/inhibit cytokinesis' 1. and 3. Accept 'prevents inhibition of cell death' for 'causes cell death' 1, 2 and 3. Accept 'cell replication' OR mitosis for cytokinesis 3. Accept graph readings >2000 to 8000 for highest

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08.4	Correct answer in range 19.3 to 19.7 = 2 marks;; Accept for 1 mark evidence of 70 and 2000 (correct readings from the graph) / 30 and 2000 (correct readings from the graph) OR 1930 (correct increase in MiTMAB, 70 to 2000) / 1970 (correct increase in MiTMAB, 30 to 2000) OR Division by 100/multiplication by 0.01 (correct conversion to mass in 0.01 dm³);	2 (2 x AO2)	Correct answer 19.3 (is obtained from 70 and 2000) 19.7 (is obtained from 30 and 2000) Accept for 1 mark, any value in range 30 to 70 and 2000 Accept for 1 mark, any value in the range 1930 to 1970
